



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination (resit)

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. You may freely utilise all results proven in the lecture. Justify clearly your answers and please write your anonymous code at the top of each sheet.

1. (a) Give an example of an infinite group G such that every non-free transitive G -set is finite. (2 pts.)
(b) Let G be a group of order 21. Prove that a G -set X of cardinality 11 must have at least one fixed point. (3 pts.)
2. Let $p > q$ be prime numbers. Prove that a group of G order $p^n q$, $n \geq 0$, contains a unique normal subgroup $P \trianglelefteq G$ such that $[G : P] = q$. (5 pts.)
3. Let R be a ring and suppose given a commutative square of R -module homomorphisms (indicated with solid arrows below) (5 pts.)

$$\begin{array}{ccccc} \ker f & \xrightarrow{i} & M & \xrightarrow{f} & N \\ \downarrow \alpha & & \downarrow v & & \downarrow u \\ \ker g & \xrightarrow{j} & M' & \xrightarrow{g} & N' \end{array} \quad u \circ f = g \circ v$$

where i and j are the canonical inclusions of the respective kernels. Suppose that

$$\ker f \cap \ker v = \{0\} \quad \text{and} \quad \ker g = \text{im}(v \circ i).$$

Prove that there exists a unique R -module homomorphism $\alpha: \ker f \rightarrow \ker g$ rendering the above diagram commutative (indicated with a dashed arrow above) and, moreover, that α is an isomorphism of R -modules.

4. Consider the following subset of the ring $\text{Mat}_{3 \times 3}(\mathbf{k})$ of 3×3 matrices with entries in a field $\mathbf{k}$:

$$R := \left\{ \begin{pmatrix} a & b & c \\ 0 & d & 0 \\ 0 & 0 & e \end{pmatrix} \in \text{Mat}_{3 \times 3}(\mathbf{k}) \mid a, b, c, d, e \in \mathbf{k} \right\}.$$

- (a) Prove that R is a (unital) subring of the ring $\text{Mat}_{3 \times 3}(\mathbf{k})$. (1 pt.)
- (b) Prove that the regular representation of R admits a direct-sum decomposition (4 pts.)

$$R \cong P_1 \oplus P_2 \oplus P_3$$

where P_1, P_2, P_3 are pairwise non-isomorphic projective R -modules.

5. Let R be an integral domain that is also an Artinian ring. Prove that R is a field. (5 pts.)
6. Determine the elementary divisors and the invariant factors of the abelian group (5 pts.)

$$(\mathbb{Z}^2 \oplus \mathbb{Z}/9\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/3\mathbb{Z})^2.$$